

EXPERIMENT NO. 4 / 5

Scrum Activity Log & JIRA Agile Management

Project: Hospital Management System (HMS)

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1. Aim

To plan, track, and manage the Hospital Management System (HMS) project using the Agile Scrum framework and JIRA Software, organizing deliverables into 5 sprints with prioritized user stories, story points, and burndown tracking.

2. Scrum Setup & Team Roles

Scrum Role	Assigned Member	Core Function in Project
Product Owner	Aaryan Choube (24CE1045)	Defines Epics, prioritizes Product Backlog, ensures clinical requirements compliance.
Scrum Master	Anish Chitnis (24CE1044)	Facilitates Scrum events, removes technical blockers, tracks sprint velocity in JIRA.
Development Team	Somansh Dafade & Team	Full-stack development (Spring/React/DB), unit testing, code reviews, and CI/CD.

3. Sprint Cadence (5 Sprints - Total 47 Days)

Sprint	Focus Area / Goal	Story Points	Duration	Key Deliverables
Sprint 1	Auth, DB Schema & Patient Reg	18 pts	Days 1–9	Patient registration form, RBAC user authentication, MySQL schema setup.
Sprint 2	Doctor Schedule & Appointment	21 pts	Days 10–20	Doctor slot configuration, token issuance, booking calendar UI.
Sprint 3	Consultation, EHR & Lab Module	19 pts	Days 21–29	Clinical notes capture, e-prescription generation, lab order upload.

Sprint 4	Billing, Insurance & Payment	23 pts	Days 30–38	calculateBill() implementation, insurance discount rule, UPI payment gateway.
Sprint 5	Integration, Security & UAT	15 pts	Days 39–47	White-box basis path testing, penetration testing, final deployment.

4. Sample User Stories (Product Backlog)

- HMS-STORY-1: As a Receptionist, I want to quickly register a new walk-in patient so that an auto-generated Patient ID is issued within 30 seconds. (Story Points: 3, Priority: Highest)
- HMS-STORY-2: As a Patient, I want to view available doctor time slots so that I can schedule an appointment without queuing. (Story Points: 5, Priority: High)
- HMS-STORY-3: As a Doctor, I want to prescribe medications from the integrated formulary so that pharmacy stock is automatically checked. (Story Points: 5, Priority: High)
- HMS-STORY-4: As a Cashier/Patient, I want the system to calculate the net bill applying insurance discounts so that payment can be processed digitally. (Story Points: 8, Priority: Highest)

5. JIRA Board & Agile Ceremonies

- JIRA Workflow Columns: To Do → In Progress → Code Review → Testing → Done.
- Sprint Planning: Held at the start of each sprint to select backlog user stories and estimate story points using Planning Poker.
- Daily Standup: 15-minute daily sync discussing: What was accomplished yesterday? What is planned today? Are there any blockers?
- Sprint Review & Demo: Working software increment demonstrated to stakeholders at the end of each sprint.
- Sprint Retrospective: Team evaluates process efficiency, identifying improvements for subsequent sprints.

6. Conclusion

By implementing Agile Scrum and tracking tasks in JIRA Software, the HMS project achieved high team transparency, rapid iterative feedback, and predictable delivery across all five development sprints.